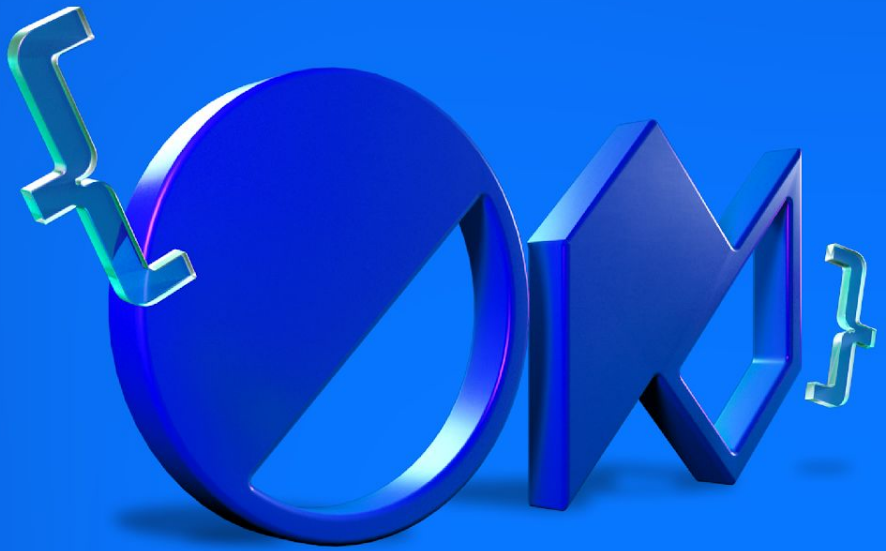
What Took Us Weeks in 2018, Takes Minutes in 2026: Searching Millions of Images in Elasticsearch

Elastic's role in image classification pipelines

Aakash Dhakshnamoorthy and Navin · Elastic



elastic

What we'll cover

01	**Where the pipeline started**	img2vec for embeddings, Elastiknn for nearest-neighbor search
02	**What ran at scale**	100 million+ images vectorized and searched in Elasticsearch
03	**What Elastic added, release by release**	From approximate kNN in 8.0 to bfloat16 vectors in 9.3
04	**The Elastic-native pipeline today**	Inference endpoints, quantized vector fields, semantic query
05	**From semantic search to agents**	Agent Builder on the same retrieval foundation

The core message

BEFORE

External tools for vectors, plus orchestration you build and operate.

Three systems to run: a model server, a vector store, a search plugin.

AFTER

Elastic is the engine for ingestion, vectorization and retrieval.

One platform: inference endpoints, vector fields, semantic query.

Before: four stages, three of them mine to run

SOURCE

Image corpus

Product and catalog imagery

VECTORIZE

img2vec

Pre-trained PyTorch vision models

STORE

Elasticsearch

Dense float and sparse boolean vectors

SEARCH

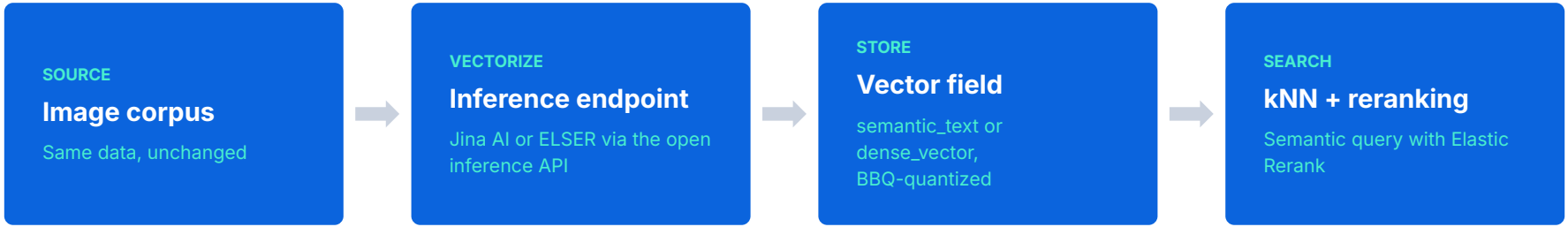
Elastiknn plugin

Exact and approximate kNN

Where the operational work sat

- **Integration stack**
  Extension and plugin install on every cluster
- **Model operations**
  Serving, versioning and scaling the embedding model
- **Orchestration**
  Glue code between vectorization, indexing and search

After: one platform, four managed stages



What that removed

- **No model serving**
  Inference endpoints replace the custom ML pipeline
- **No plugin to maintain**
  kNN is native to Elasticsearch from 8.0 onward
- **Managed scaling**
  Elastic Cloud Serverless handles capacity

Sources: open inference API and semantic_text documentation, elastic.co/docs

Previous library: Elastiknn

How I used it

- Nearest-neighbor search before Elastic-native vector search existed.
- Supported both dense float vectors and sparse boolean vectors.
- Ran similarity search with exact and approximate approaches.

CREDIT

Alex Klibisz

Special thanks for building and maintaining Elastiknn.

github.com/alexklibisz/elastiknn

Elasticsearch plugin for similarity / nearest-neighbor search.

Previous library: img2vec

How I used it

- Generated image embeddings before the Elastic-native inference flow.
- Used pre-trained PyTorch vision models for feature extraction.
- Indexed the output vectors for similarity and semantic-style retrieval.

CREDIT

Christian Safka

Special thanks for building and sharing img2vec.

github.com/christiansafka/img2vec

Use pre-trained PyTorch models to extract image embeddings.

Scale and outcome

From the previous pipeline

100 million+

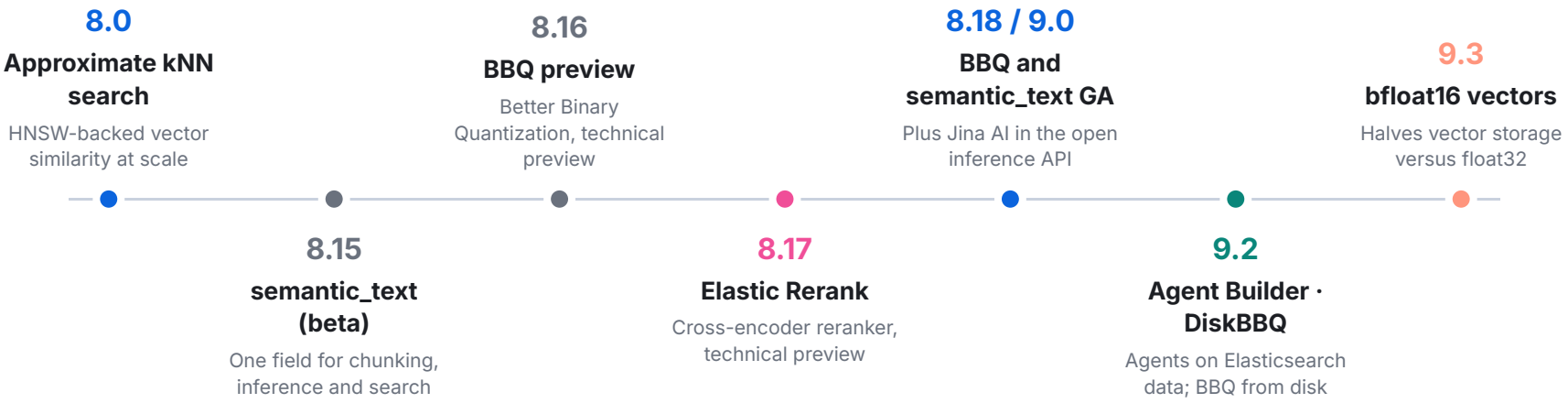
images converted into vectors and indexed in Elasticsearch

What I built

- img2vec for image vectorization.
- Elastiknn for nearest-neighbor similarity search.
- 100 million+ images converted into vectors.
- Vectorized data injected into Elasticsearch.

Result: semantic search stayed fast at that volume — which is what justified rebuilding it natively on Elastic.

How Elastic closed the gap, release by release

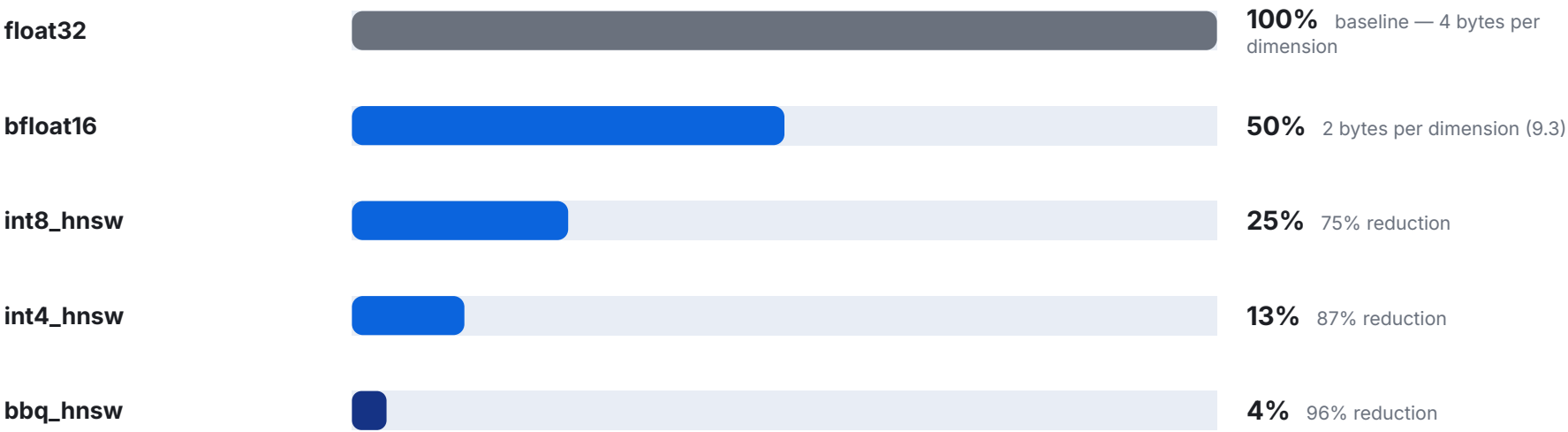


The pattern: every capability the old stack needed an external component for has become a native Elasticsearch feature.

Sources: Elastic release blogs for 8.0, 8.15, 8.16, 8.17, 8.18/9.0 and 9.2; bfloat16 announcement on Elasticsearch Labs.

Why this scales now: vector storage economics

Relative memory footprint of one float vector, by index type



From 9.0, **bbq_hnsw** is the default index type for float vectors of 384 dimfloat32: 1024 × 4 bytes = 4,096 bytes per vector → 409.6 GB BBQ: 1024 × 1 bit = 128 bytes per vector → 12.8 GB (32× smaller) ensions or more, and **int8_hnsw** below that.

float32: 1024 × 4 bytes = 4,096 bytes per vector → 409.6 GB
BBQ: 1024 × 1 bit = 128 bytes per vector → 12.8 GB

Elastic Agent Builder

What it gives you

- A built-in conversational agent over your Elasticsearch data.
- Hybrid search relevance through built-in ES|QL or custom tools.
- Workflows (preview) as reusable tools, plus MCP and A2A integration.

Why it matters for this talk

- The same retrieval foundation used for image and vector search.
- The natural next step from semantic search to agentic applications.
- Generally available from 9.2, on Serverless and the Enterprise tier.

Source: Elastic 9.2 release blog and the Agent Builder GA announcement, elastic.co

Agent capabilities on the same index

Knowledge retrieval

Hybrid and semantic search over indexed content

Find visually similar products from a text prompt

Grounded responses

Context from vectors, keywords and metadata filters

Answers tied to exact catalog documents

Reranking

Elastic Rerank reports ~40% average relevance gain over BM25

Better top-k ordering before the final answer

Production scale

Serverless deployment with managed scaling

High-volume image and multimodal search agents

Reranking figure: Elastic Rerank model documentation, averaged across 21 datasets.

Elastic references

Approximate kNN in 8.0
elastic.co/blog/introducing-approximate-nearest-neighbor-search...

dense_vector and quantization
elastic.co/docs/reference/elasticsearch/mapping-reference/dense-vector

Elastic Inference Service
elastic.co/docs/explore-analyze/elastic-inference/eis

Jina endpoint API
elastic.co/docs/api/.../operation-inference-put-jinaai

Semantic search quickstart
elastic.co/docs/solutions/search/get-started/semantic-search

Elastic Cloud Serverless
elastic.co/docs/deploy-manage/deploy/elastic-cloud/serverless

Project links

My repositories

kNN Image Search
github.com/navinbalaji/elastic-knn-image-search

Elastic Copilot Search
github.com/navinbalaji/elastic-copliot-search

External libraries

img2vec
github.com/christiansafka/img2vec

Elastiknn
github.com/alexklibisz/elastiknn

Elastic Labs reading

Better Binary Quantization
elastic.co/search-labs/blog/better-binary-quantization-lucene-elasticsearch

bfloat16 vectors in 9.3
elastic.co/search-labs/blog/bfloat16-vector-support-elasticsearch

Agent Builder GA
elastic.co/search-labs/blog/agent-builder-elastic-ga

Thank you!

Navin Balaji

Questions welcome

